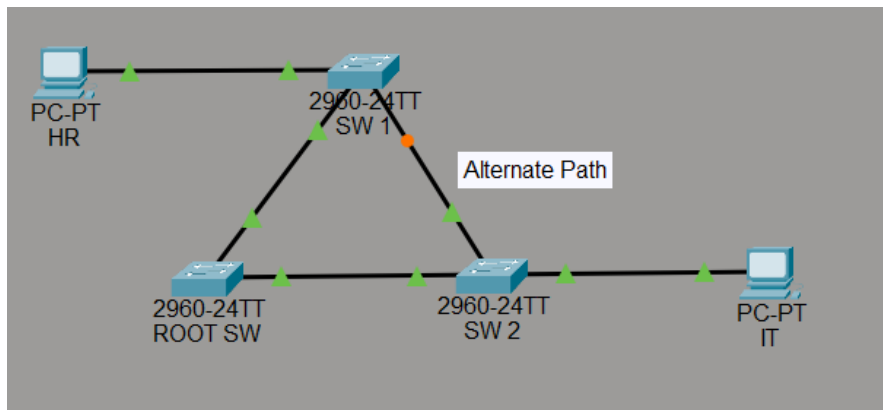


MINI PROJECT 4

Redundant Layer-2 Topology with Rapid PVST+ (STP) + VLANs

DIAGRAM:



DESCRIPTION:

This project shows a redundant Layer-2 network topology using VLANs and Rapid PVST+ (STP) to prevent switching loops. It explains how STP blocks and unblocks ports to keep the network stable while still providing backup paths.

STEP 1: Create VLANs on SW 1,SW2,ROOT SWITCH:

Verify existing VLANs from Previous projects

STEP 2: Assign Access Ports to VLANs on SW1 & SW2:

SW1:

```
interface fastEthernet0/1
switchport mode access
switchport access vlan 10
exit
```

SW 2:

```
interface fastEthernet0/1
switchport mode access
switchport access vlan 20
exit
```

STEP 2: Configure Trunk Links Between Switches

```
interface range fastEthernet0/22 - 24
switchport mode trunk
switchport trunk allowed vlan 10,20
exit
```

STEP 4: Configure IP Addressing on PCs:

PC	VLAN	IP Address	Gateway
PC-HR	10	192.168.10.10/24	192.168.10.1
PC-IT	20	192.168.20.10/24	192.168.20.1

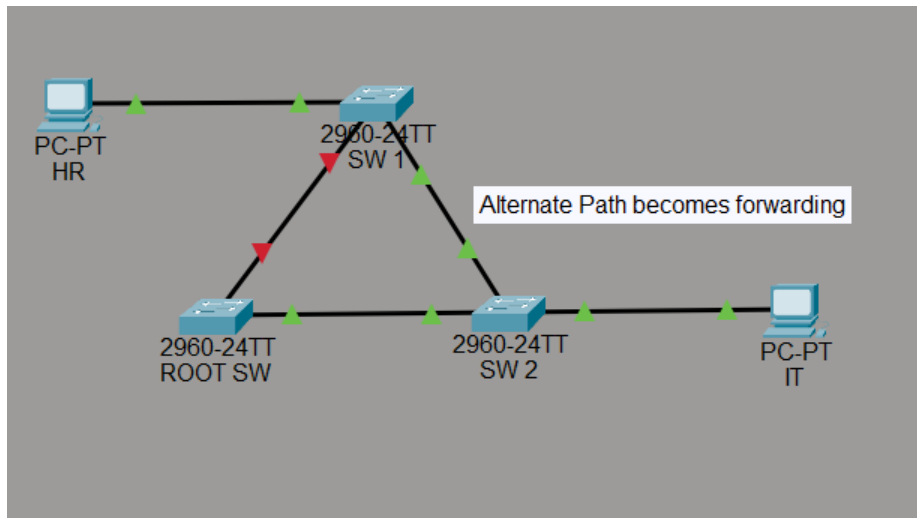
STEP 5: Enable Rapid PVST+(default in 2960 cisco swtich):

```
enable
configure terminal
spanning-tree mode rapid-pvst
exit
```

STEP 6: Observe Port Roles & Blocking:

```
show spanning-tree
```

STEP 7: Failure Simulation (Critical Behavior) & Ping Test:



we do not have Router or Layer 3 switch, so ping test is not possible between two networks.